PRANAVA PREETHIVARDHAN CHANDURI

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EDUCATION

Master of Science - Data Science 05/27
University Of North Texas GPA: 4.0/4.0
Relevant Coursework: Deep Learning, Natural Language Processing, Machine Learning, Software Development for AI, Data Modeling

Bachelor of Technology - Computer Science and Engineering Specialization in AI & ML 05/25
SRM University GPA: 3.19/ 4.0
Relevant Coursework: Computer Vision, Image Processing, Operating system, Computer Architecture, Data Structures, Databases

SKILLS

Programming Languages: Python, C++, SQL, R (Basic)
Software Engineering: Data Structures, Algorithms, Object-Oriented Programming, System Design, API Development , Microservices , Distributed Systems, Secure Coding Practices
Backend & Databases: REST APIs, MySQL, PostgreSQL, Database Optimization, Query Performance Tuning
Cloud & DevOps: AWS EC2, Docker, CI/CD Pipelines, Git, Linux, Containerized Applications
Data & ML Engineering: ETL Pipelines, Data Processing, Parallel Computing, Machine Learning, Model Evaluation, PyTorch
Certifications: Machine Learning & AI (LinkedIn Learning), Prompt Engineering with ChatGPT, Unity and Vuforia in Augmented Reality

PROFESSIONAL EXPERIENCE

Graduate Research Assistant | Denton, TX 08/25 – 03/26
Data Visualization & Extreme Reality Lab, University of North Texas

1. Contributed to the development of an immersive Virtual Reality (VR) and Mixed Reality (MR) system to enhance Cyber Situational Awareness (CSA) for real-time threat detection and response.
2. Built and visualized 3D Force-Directed Node Graphs in Unity 3D using cybersecurity datasets (CTU-13, CICIDS, NSL-KDD), rendering network topologies, traffic flows, and intrusion paths on Meta Quest 3 and Microsoft HoloLens 2.
3. Parsed and loaded large-scale network datasets (CSV/JSON/XML) into Unity using C# scripts and UnityWebRequest, enabling real-time animated attack visualizations and dynamic 3D object instantiation.
4. Developed interactive VR modules featuring voice-controlled threat filtering, hand-gesture interaction, and animated attack timelines to reduce cognitive load for cybersecurity analysts.
5. Collaborated in a cross-institutional research team (UNT + GPREC) applying Agile research workflows across a 14-month multi-phase development lifecycle.

Software Developer, Intern | MSRcosmos, Hyderabad, India 06/23 – 08/23

1. Developed and deployed a full-stack e-commerce mobile application with product listings, shopping cart, user authentication, and order management, integrating RESTful APIs for seamless high-volume transactional data exchange.
2. Designed modular, reusable UI components with structured exception handling and input validation, while optimizing SQL queries and indexing strategies to reduce database latency by 25%.
3. Integrated the application into CI/CD pipelines with automated testing workflows, ensuring production-grade build quality and consistent releases.
4. Collaborated in an Agile team across sprint planning, code reviews, debugging, and defect resolution to deliver reliable software on schedule.

PROJECTS

Satellite Image Land Classification LINK **(Python, TensorFlow, Flask, OpenCV)**

1. Built a deep learning-based satellite image land classification and segmentation system using the DeepGlobe Land Cover Classification dataset to identify urban, vegetation, water, barren land, and agricultural regions.
2. Developed an end-to-end pipeline for preprocessing, augmentation, model inference, post-processing, and visualization, using techniques such as resizing, normalization, flipping, rotation, and contrast enhancement.
3. Created a Flask web application to upload satellite images, generate segmented land-cover maps, and display land distribution percentages, evaluated with IoU, Precision, Recall, F1-score, and confusion matrices.

Demand Forecasting ML Pipeline LINK **(Scikit-learn, XGBoost, LSTM, Time-Series Modeling)**

1. Built an end-to-end machine learning pipeline to forecast daily retail demand across product store time series using gradient boosted trees.
2. Engineered high-signal time series features including lag effects, rolling statistics, seasonality, price trends, event indicators, and state-level SNAP factors.
3. Implemented leakage-free temporal validation and trained an optimized XGBoost model with early stopping, outperforming naïve baseline forecasts.
4. Designed a production-style modular architecture with reproducible pipelines, CLI training/inference workflows, and automated 28-day iterative forecasting.

Agentic RAG System LINK **(Python, FastAPI, FAISS, Sentence Transformers, LangChain)**

1. Built a hybrid retrieval system combining BM25 and FAISS vector search with Reciprocal Rank Fusion, cutting down irrelevant context fed to the LLM by dynamically routing queries through an agentic planner.
2. Boosted answer accuracy by stacking a Cross-Encoder reranking layer with a citation validator, ensuring every generated response is backed by traceable source chunks rather than hallucinated content.
3. Shipped the full RAG pipeline as a production REST API using FastAPI, paired with a Streamlit UI for non-technical users, following a clean modular codebase structure ready for team collaboration.
4. Prevented silent performance drops by wiring up an evaluation suite tracking Recall@K, MRR, and nDCG@K, making it easy to catch retrieval regressions before they hit production.